

CHAPTER 1 RESULTS

1.1 Cohort Characteristics

TCGA-LUSC datasets for five modalities were retrieved (described in Section ??).

Gene expression and Copy number variation (CNV) retained the full labelled 178-sample cohort, whereas Deoxyribonucleic acid (DNA) methylation and miRNA expression had smaller assay overlap (Table 1.1). Across modalities, the classical subtype was the most frequent class and primitive the least frequent, motivating the use of macro-averaged classification metrics in subsequent analyses.

Table 1.1: Summary of omics modalities used in this study. Variation in sample size reflects assay availability and label overlap.

Modality	Configuration	Classical	Basal	Secretory	Primitive	Total
Gene expression	HiSeqV2, RNASeqV2 RSEM-normalised	65	43	43	27	178
DNA methylation	HumanMethylation450k beta values	29	14	22	11	76
miRNA expression	miRNA sequencing, log2(RPM + 1)	44	29	20	14	107
RPPA	Normalised protein abundance	38	28	28	18	112
CNV	GISTIC2 all-thresholded calls	65	43	43	27	178

1.2 Preprocessing Summary

Table 1.2: Preprocessing and feature-retention summary for each omics modality. Raw feature counts correspond to the input assay feature spaces. The MAD threshold is the score of the last retained feature in the processed matrix; for RPPA, where all 238 antibodies were retained, this value represents the minimum observed MAD among retained antibodies.

Modality	Raw features	Retained features	MAD threshold
Gene expression	60,660	2,000	1.0808
DNA methylation	485,577	2,000	0.2208
micro RNA (miRNA) expression	1,881	200	0.5954
RPPA	238	238	0.0692
CNV	24,776	2,000	1.0000

1.3 Characterization of LUSC Subtypes

1.3.1 Clinical Characteristics

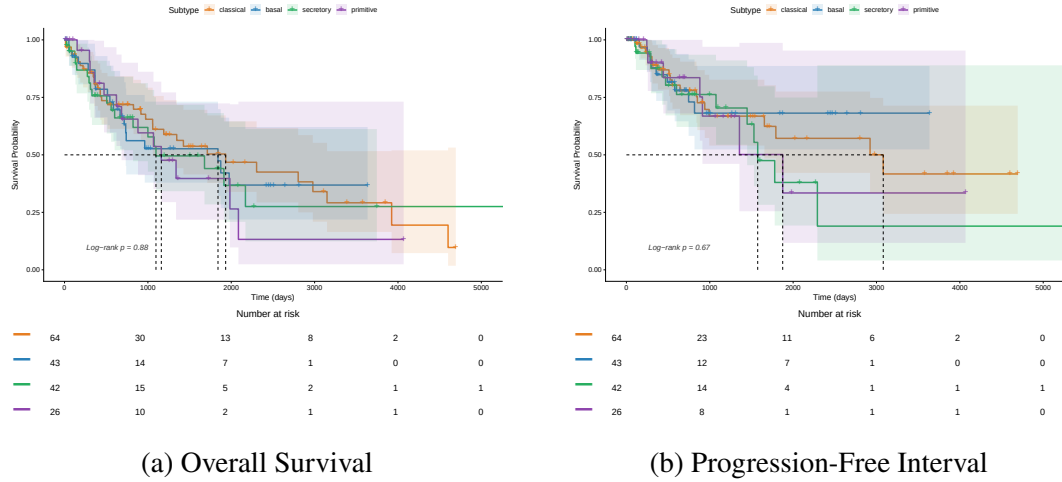


Figure 1.1: Survival curves showing Overall Survival (a) and Progression-Free Interval (b) for TCGA-LUSC samples (n=178) stratified by molecular subtypes. Median survival and 95% confidence intervals were computed by the Brookmeyer-Crowley method. Shaded regions denote 95% confidence intervals and risk table shows at-risk patient counts at selected time points.

Secretory and primitive subtype show the shortest median OS of 36 months and 38.1 months respectively, consistent with known proliferative and poorly differentiated biology of primitive subtypes (Hammerman et al., 2012; Wilkerson et al., 2010).

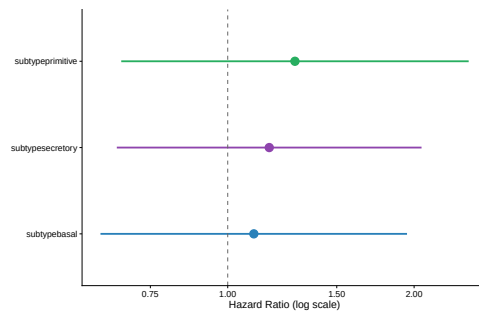


Figure 1.2: Hazard ratios for overall survival by LUSC molecular subtype from Cox proportional hazards regression, with classical subtype as reference (n=AVOCADO). Points denote hazard ratios and horizontal bars represent 95% confidence intervals. HR > 1 indicates worse prognosis relative to classical subtype. The proportional hazards assumption was verified using scaled Schoenfeld residuals (cox.zph global p=AVOCADO). Concordance index=AVOCADO

- 1.3.2 Immune Microenvironment**
- 1.3.3 Somatic Mutation Landscape**
- 1.4 Dataset Separability**
- 1.5 Performance Across Omics Modalities**
- 1.6 SHAP Feature Importance and Stability**
- 1.7 Cross-Model Feature Overlap**
- 1.8 Cross-Modality Feature Overlap**
- 1.9 Pathway Enrichment**

References

- Hammerman, P. S., Lawrence, M. S., Voet, D., Jing, R., Cibulskis, K., Sivachenko, A., Stojanov, P., McKenna, A., Lander, E. S., Gabriel, S., Getz, G., Sougnez, C., Imielinski, M., Helman, E., Hernandez, B., Pho, N. H., Meyerson, M., Chu, A., Chun, H.-J. E., ... group, D. w. (2012). Comprehensive genomic characterization of squamous cell lung cancers. *Nature*, 489(7417), 519–525. <https://doi.org/10.1038/nature11404>
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